Problem Set 1 (QBCS 5002: Introduction to Quantitative Biology)

Due: September 17, 2026, at 5:00 p.m.

This problem set is to be completed by teams of two. We strongly encourage teams pairing a student from a biology, medicine or chemistry background with a student from a mathematics, physics or engineering background. Each team will submit one joint solution, and both partners will receive the same grade.

Both partners should be prepared to explain and modify any part of their submitted reasoning in class after the deadline. Selected teams may be invited to do so; not every team will necessarily be called upon.

This problem set is being released after the lecture on September 4th so that you can begin forming teams, selecting questions and constructing divide-and-conquer diagrams. The lecture on September 8th will introduce additional ideas about bacterial growth, the central dogma and rates that will be useful for Problem 1. You are not expected to complete that problem before September 8th.

The complete submission should not exceed eight pages, excluding references and diagrams.

Use of generative AI**:** This problem set is designed for practicing thinking tools learnt in the course. Generative-AI tools may therefore not be used to formulate, solve or write the solutions. You may use course materials, reference books, calculators and search engines.

Problem 1: Ribosome traffic during rapid growth

A rapidly growing *E. coli* cell can double in approximately 20 minutes. Under these conditions, several ribosomes can translate the same mRNA simultaneously.

Estimate:

- How much time can one ribosome afford to spend adding one amino acid?
- How frequently must translation be initiated on an average mRNA for the cell to double its protein content in 20 minutes? Express your answer both as an initiation rate and as the average time between successive initiation events.
- What average separation between neighboring ribosomes do these two rates imply? Express the separation in codons or nucleotides and compare it with an order-of-magnitude estimate of the length of mRNA physically occupied by one ribosome.

Use the following anchor numbers:

- The protein:total-RNA mass ratio is approximately 2:1. (BioNumbers 103890).
- Of the total RNA, approximately 80–85% is rRNA, 10–15% is tRNA and 2–4% is mRNA. (BioNumbers 102342, BioNumbers 102343).
- An *E. coli* ribosome is approximately 20-25nm across and is approximately two-thirds RNA and one-third protein by mass. (BioNumbers 102320, BioNumbers 100119)

Construct a model connecting the amount of protein that must be produced during one generation to the operation of individual ribosomes and individual mRNAs. Decide which additional quantities must be estimated first and how they should be connected.

Try to use only the supplied anchor numbers or anchor numbers introduced in class. You will be evaluated on the economy and justification of any additional external anchors and on the clarity of your chains of reasoning—not on the accuracy of the final numerical results.

Problem 2: A BioNumbers curator's note

Write a curator's note for a BioNumbers entry:

Choose one of the following two entry points:

A. An independent route

Choose a BioNumbers entry and develop a way to estimate it that differs substantially from the measurement or estimation method used in the cited original paper. Compare the two routes and explain what assumptions they share. For the chosen entry, it is required to compare your method with the original paper, not just the webpage of BioNumbers.

B. A missing number

Identify a biologically meaningful quantity that you cannot find in BioNumbers. Explain why it would be useful, provide a defensible order-of-magnitude estimate or range, and propose an experimental strategy for measuring it.

Problem 3: A Fermi problem from your research

Choose a quantitative question related to your current, previous or prospective research—or to a biological problem you would seriously consider studying.

Your submission should:

- State the question and explain why its answer matters to the research.
- Construct a divide-and-conquer diagram.
- Use anchor numbers minimally and explicitly state their source.
- Produce an order-of-magnitude estimate with units.
- Find the best available experimental or literature value, compare your estimate with that value, and discuss what the agreement or discrepancy teaches you.

You will be evaluated on the clarity of your chains of reasoning, the originality of the scientific question you raised and the ingenuity of your devised solution —not on the accuracy of the final numerical results.